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Electricity trade impacts on regional power pools in Sub-Saharan Africa

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ABSTRACT

In this study, we evaluated the impacts of 3 different electricity trade scenarios among four African power pools given a time series of available hydropower production. The available hydropower production is determined after all other higher priority water demands are met using a river basin model (WEAP). The objective of the power pool model (RTEP) was to maximize global welfare (i.e., meet total electricity demand at lowest cost including the cost of non-served energy) assuming perfect competition.

From the WEAP analysis, the CAPP produces the most hydropower followed by the EAPP and SAPP and lastly the WAPP. This is also seen in the RTEP analysis which shows the connection of the two models. In the No Trade and Min Trade scenarios, there is a significant amount of hydropower production that goes unused in all regions, especially in the CAPP and SAPP. A positive outcome of the Full Trade scenario is that all hydropower is used, none is wasted.

By allowing trade among the power pools, the need to build new electricity production facilities is decreased by about 19 GW in the Min Trade scenario and an additional 10 GW in the Full Trade scenario. By reducing the need to expand production capacity, these regions save money by not needing to build new facilities but instead are able to use electricity from other power pools.

Introduction

In 2015, The World Bank funded a project titled "Enhancing the Climate Resilience of Africa's Infrastructure", (ECRAI) [2]. In this project, seven major African river basins (Nile, Congo, Niger, Zambezi, Orange, Senegal, and Volta) and four power pools (Central, Eastern, Southern, and West African) were studied. The analysis looked at the hydropower capacity enhancements and projected irrigation additions based on the Program for Infrastructure Development in Africa (PIDA); shown in Fig. 1. PIDA was endorsed in 2012 by Africa's government officials to close the infrastructure gap throughout Africa. Part of the analysis looked at the impact of climate change on these future plans.

The study presented in this paper built upon the work of the ECRAI project. Using the river basin models with hydropower production and the core structure of the power pool models from the ECRAI project, this current research expanded the power pool modeling to estimate the value of different levels of regional power sector integration in sub-Saharan Africa and evaluated how the value may change in the face of climate change.

Literature review

While some argue that large, connected power grids are inefficient both in energy delivery and cost [9] and [7], others believe grid connection is good for the sub-Saharan region. According to Rubomboras [6], cooperation between sub-Saharan Power Pools will generate economies of scale in utilizing shared resources in addition to providing a solution to dealing with energy impacts from local droughts. In addition, Musisi [4] states that interconnection of electric grids can provide countries the opportunity to purchase low-cost surplus power from across several borders.

Trotter et al. [10] reviewed 306 scientific articles on this topic, electricity planning and related implementation analysis in sub-Saharan Africa. They found most research focused on South Africa, Ghana, Kenya, Nigeria, Tanzania and Ethiopia, leaving other sub-Saharan African countries with little or no scholarly representation in electricity planning. Encouragingly, they found over 60% of the articles promoted renewable energy technologies. Trotter et al. [10] concluded that further well-developed research on energy provision in sub-Saharan Africa could help reach other goals like sustainable development, improved health care, mitigating climate change and eliminating poverty.

Increasing electrification rates for rural areas of high population density could occur at relatively low marginal cost according to Lee

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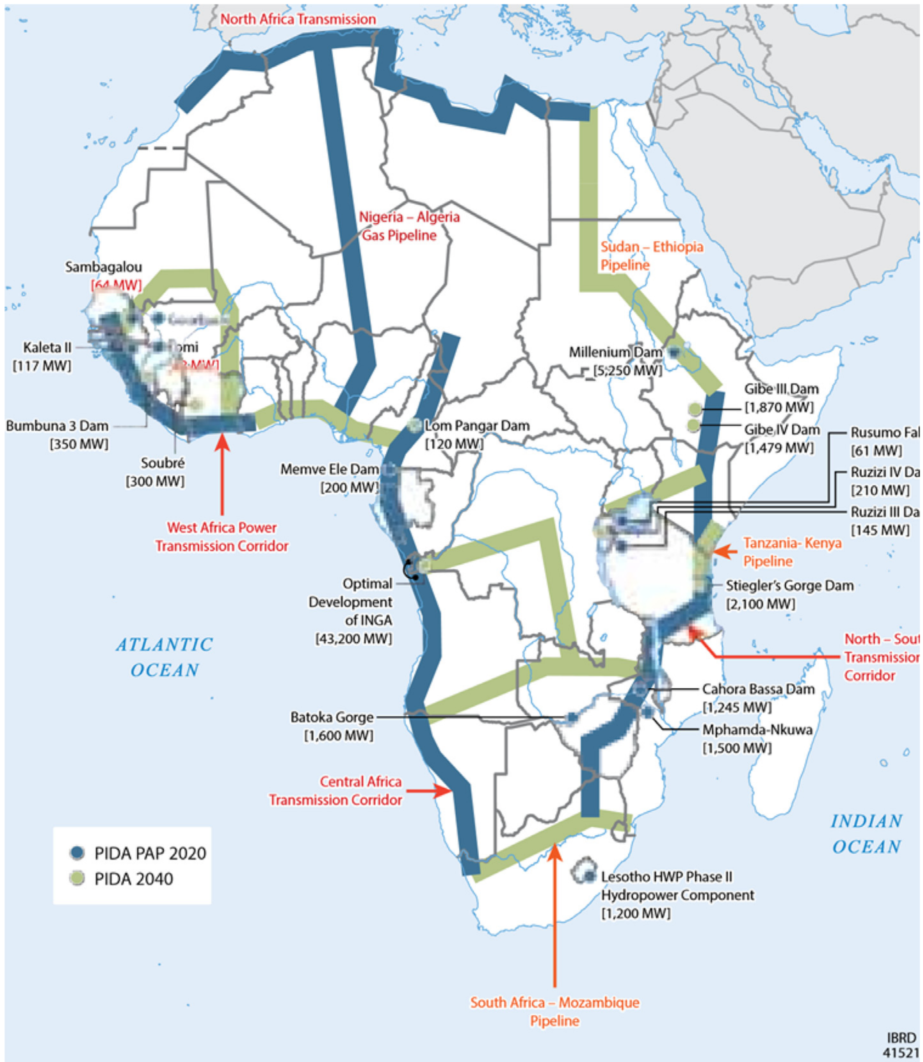

Fig. 1. Summary of PIDA Energy Infrastructure Development Proposals and Priorities, [2].

Table 1
Grouping of countries and river basins into regional power pools, [5].

	Southern African Power Pool	East African Power Pool	West African Power Pool	Central African Power Pool
Member Countries	Angola Botswana Lesotho Malawi Mozambique Namibia South Africa Swaziland Zambia Zimbabwe	Burundi Djibouti Egypt Ethiopia Kenya Rwanda Sudan[1] Tanzania Uganda	Burkina Faso Cote d'Ivoire Gambia Ghana Guinea Guinea Bissau Liberia Mali Niger Nigeria Senegal Sierra Leone Togo/Benin[2]	Cameroon Central African Republic Chad Congo Democratic Republic of Congo Equatorial Guinea Gabon
River Basin	Orange Zambezi	Nile	Volta Senegal Niger	Congo

[1] Sudan and South Sudan were represented as a single country in order to be consistent with most of the available data.
[2] Togo and Benin were also represented as a single country because they share a common utility company and most data group these countries together.

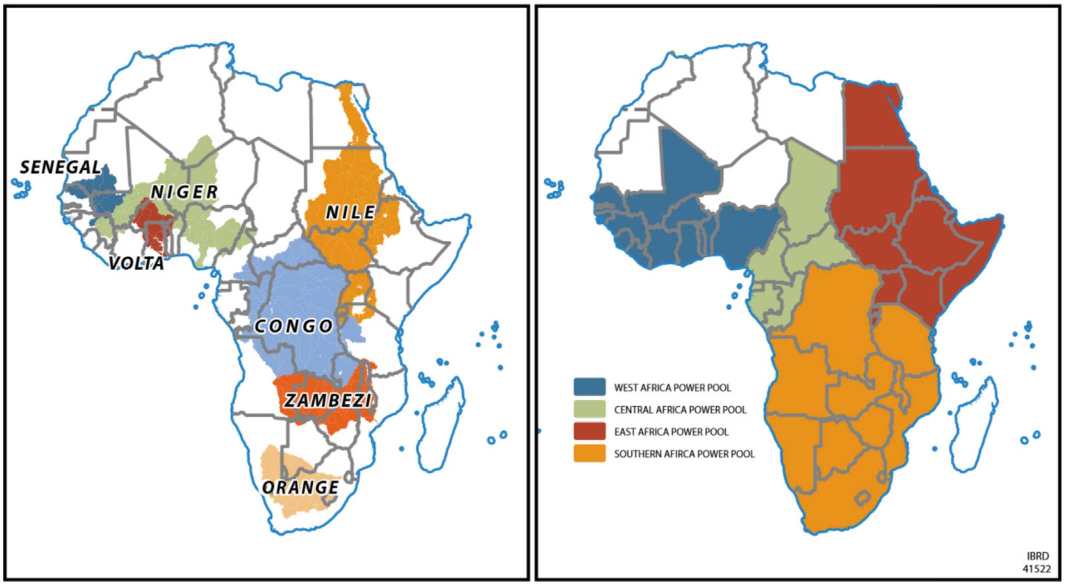

Fig. 2. Selected River Basins and Power Pools in Africa, Cervigni et. al., *[2015]*.

at al. [3]. In a case study of Kenya, Lee found many unconnected households clustered within 200m of a low-voltage line that could be connected. Considering this circumstance likely exists in other sub-Saharan African countries, an increase in connected households could occur quickly given proper policies, further increasing electricity demand in the region. Lee et al. [3] supports further research on the demand for and effects of electrification.

Bowen et. al. [1] employed a short-term cost minimizing mixed integer linear programming model in the Southern African Power Pool (SAPP) to evaluate electricity trade. The study looked at thermal and hydropower as electricity sources. The hydropower was based on water availability, which was not described in the referenced paper. It appears to be a set average availability without linking to other water demands and/or climate. The analysis used excess hydropower generation in DRC, Zambia and Mozambique to off-set more expensive coal-powered electricity plants. Bowen et al. [1] concluded potential benefits from increasing electricity trade in the SAPP to be tens of millions of dollars per year.

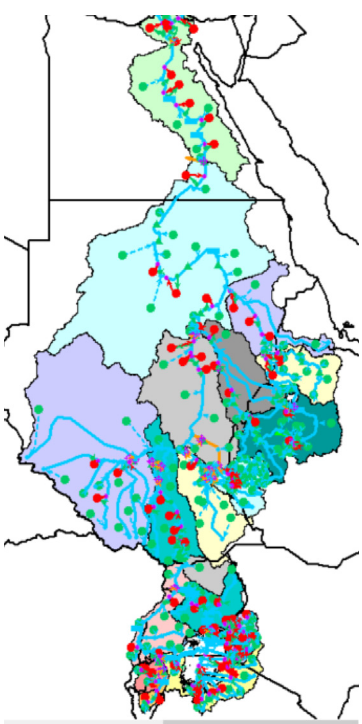

Fig. 3. Nile River Basin Representation in WEAP, Cervigni et. al., *[2015]*.

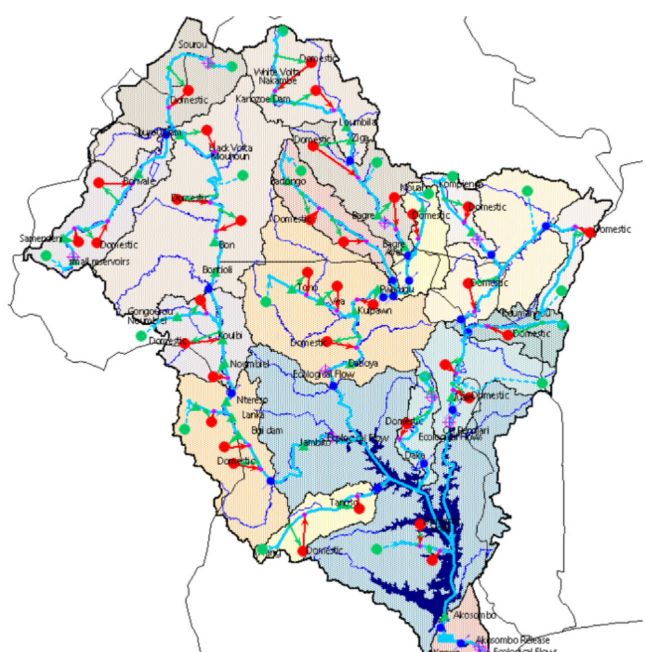

Fig. 4. Volta River Basin Representation in WEAP, Cervigni et. al., *[2015]*.

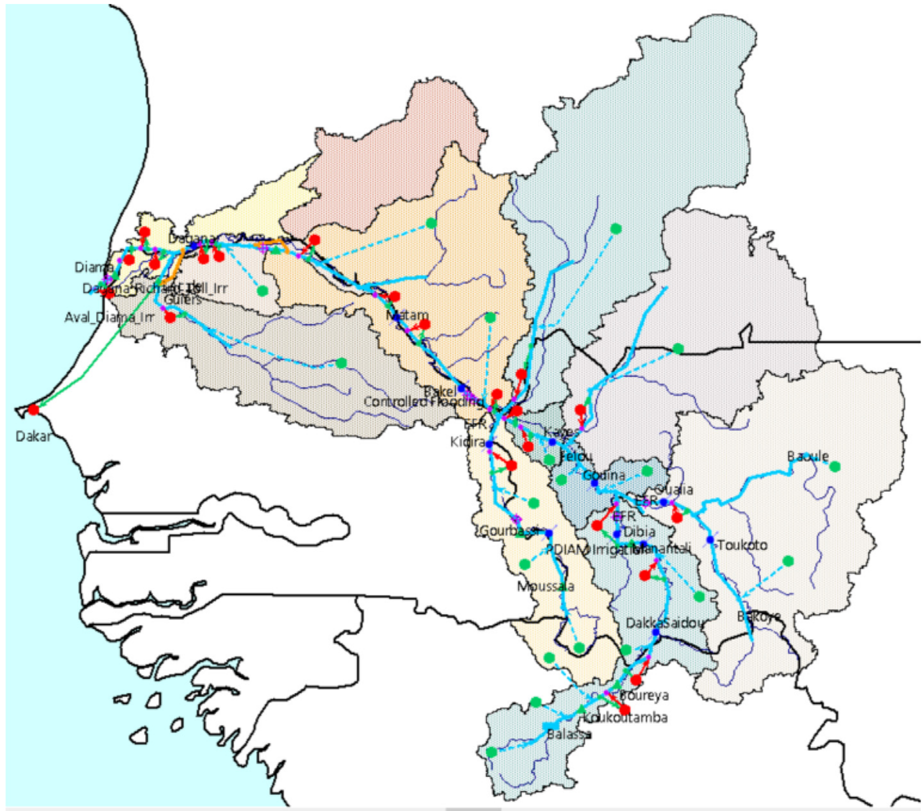

Fig. 5. Senegal River Basin Representation in WEAP, Cervigni et. al., *[2015]*.

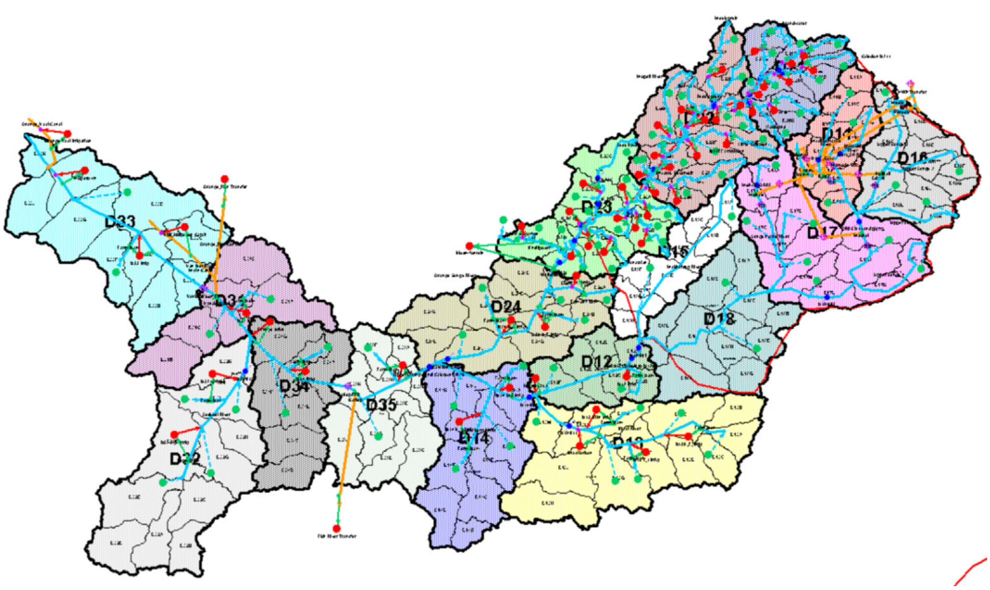

Fig. 6. Orange River Basin Represented in WEAP, Cervigni et. al., *[2015]*.

Background

Africa power pool model

The Regional Trade and Expansion Planning (RTEP) model was developed by Amy Rose and Ignacio Perez-Arriaga at MIT, [5]. This model expanded the power pool modeling efforts in the ECRAI analysis to estimate the value of different levels of regional power sector integration in sub-Saharan Africa and how this value may change in the face of climate change. For a more detailed review of the RTEP model, please see the report "Technical Note: Proposed Model for Regional Power Sector Integra-

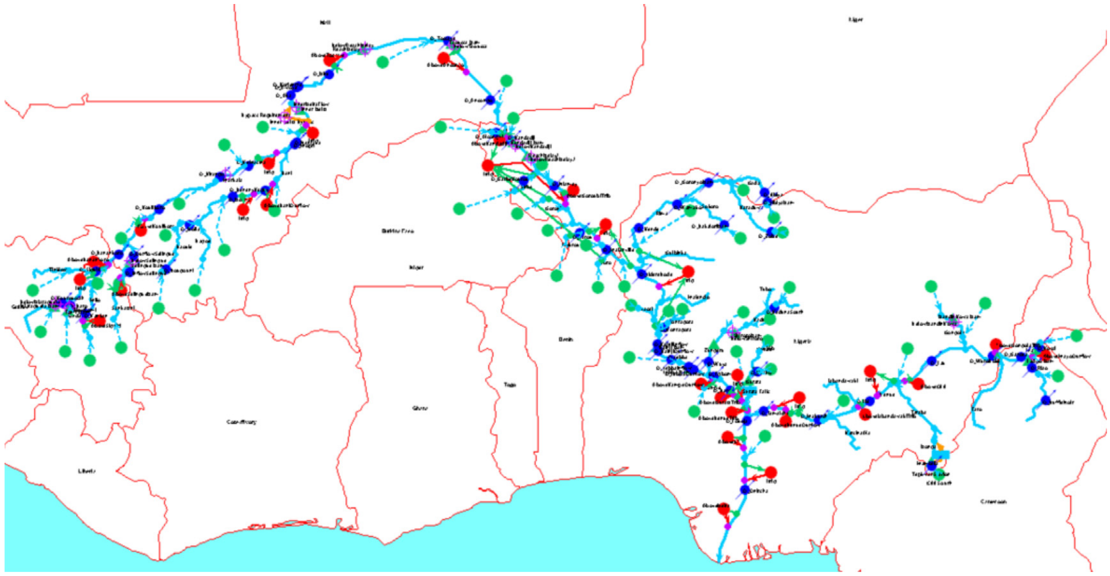

Fig. 7. Niger River Basin Representation in WEAP, Cervigni et. al., *[2015].*

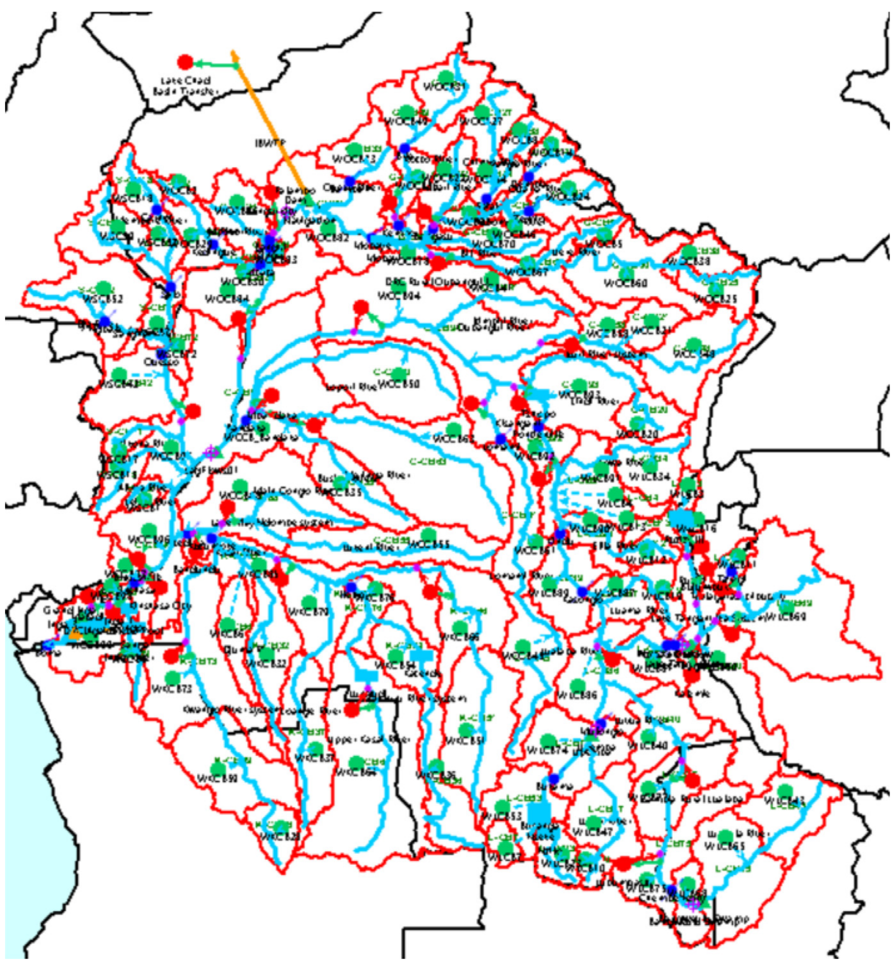

Fig. 8. Congo River Basin Represented in WEAP, Cervigni et. al., *[2015].*

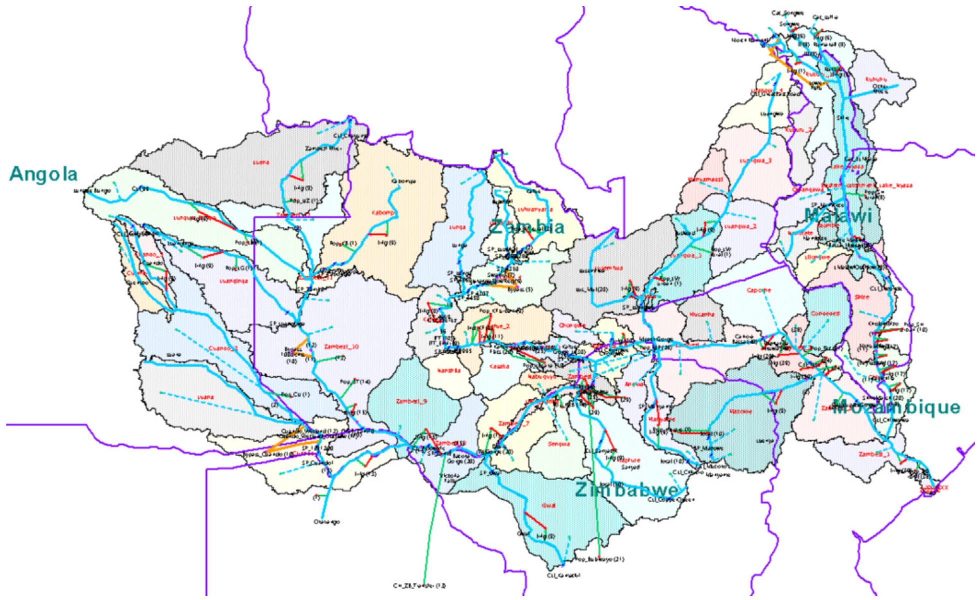

Fig. 9. Zambezi River Basin Represented in WEAP, Cervigni et. al., *[2015]*.

Table 2
WEAP country names linked to RTEP country codes.

WEAP Country	RTEP Country
Democratic Republic of Congo	DRC
Central African Republic	CAR
Zimbabwe	ZIM
Mozambique	MOZ
Zambia	ZAM
Malawi	MAL
Tanzania	TAN
Ethiopia	ETH
Egypt	EGY
Uganda	UGA
Southern Sudan	–
Sudan	–
Sudan Sum	SUD
Kenya	KEN
Nigeria	NGA
Guinea	GUI
Niger	NIG
Cameroun	CAM
Mali	MLI
South Africa	SAF
Lesotho	LES
Ghana	GHA
Burkina Faso	BFS

Note: In RTEP model Sudan contains South Sudan and Sudan

tion in Africa" [5]. The following section is a summary of the RTEP model.

The model ran from 2010 to 2030 and was subdivided into months and 3 daytime slices (peak, shoulder, and off-peak). The base electricity demand used in the analysis is provided by country in Appendix 1. The model regions, associated countries, and corresponding river basins are shown in Table 1. A visual representation of the power pool regions along with the modeled river basins is shown in Fig. 2. The RTEP model ran 3 different trade (exchanges of electricity) scenarios to evaluate the impacts of different levels of regional integration. The first scenario called the Base Case, only allowed for committed new transmission projects and had no inter-regional trade. In the Base Case, the countries supplied their own firm and operating reserves. The second scenario, called the MinTrade scenario, allowed for committed and economic transmission investments with limited inter-regional trade. In the MinTrade scenario, countries supplied their own firm and operating reserves. The third and final scenario, called the FullTrade scenario, allowed for committed and economic transmission investments with unlimited inter-regional trade. In the FullTrade scenario, countries could share reserves.

The objective of the RTEP model was to maximize global welfare (i.e., meet total demand at lowest cost including the cost of non-served energy) assuming perfect competition and considering:

- Demand projections (Appendix 2)
- Existing and committed generation and transmission infrastructure
- Resource availability
- Available generation technologies and investment costs
- Fuel prices
- Operating constraints (i.e., trade rules, security of supply, emissions policies)

The model chose the "optimal" mix of generation and transmission infrastructure.

WEAP African River Basin Representation

As part of the ECARI study, seven African river basins were modeled. These basins include Senegal, Niger, Volta, Nile, Congo, Zambezi, and Orange and are shown in Figure 2. These basins were modeled using the Water Evaluation And Planning (WEAP) model, [8]. More details on the data, model and processing can be found in the ECARI document, [2].

The WEAP model uses an integrated approach to modeling water supply, demand, infrastructure, and management. The ECARI study utilized WEAP's built-in rainfall-runoff model to determine

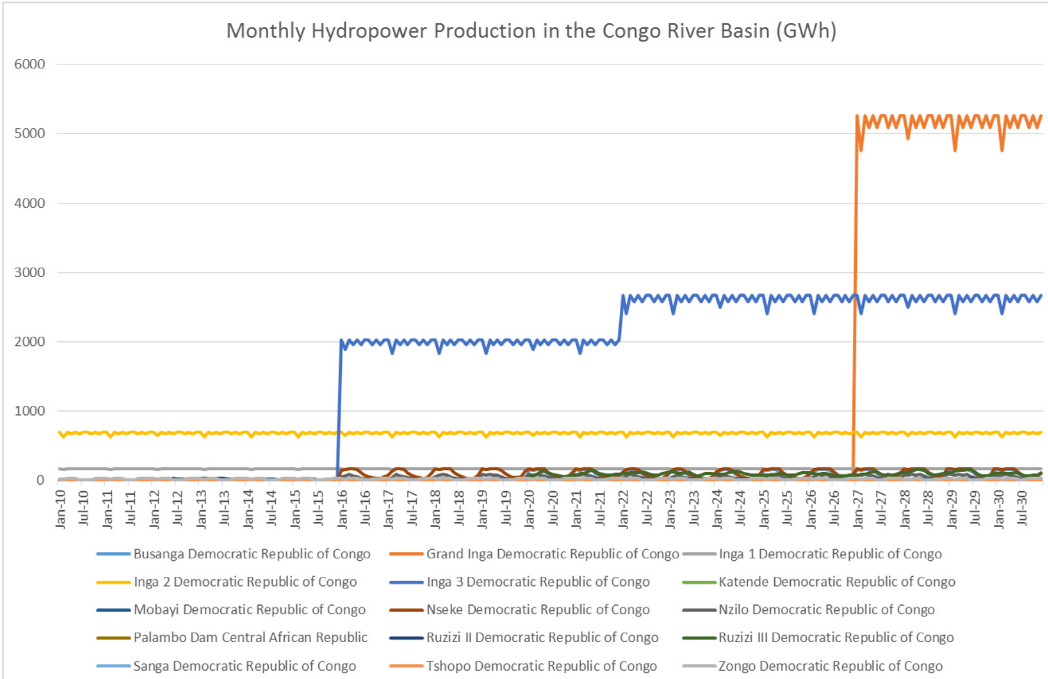

Fig. 10. Monthly Hydropower Production in the Congo River Basin (Gwh) from WEAP analysis.

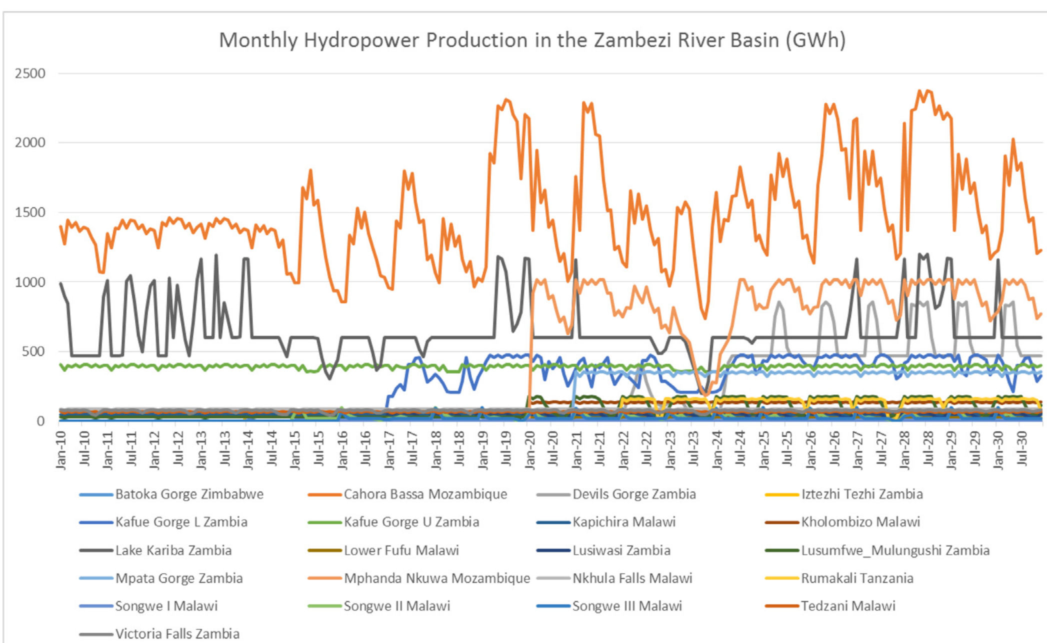

Fig. 11. Monthly Hydropower Production in the Zambezi River Basin (Gwh) from WEAP analysis.

the available runoff in each catchment. The catchment size depended on the river basin being modeled. Demands modeled included, municipal, industrial, irrigation, and environmental flow constraints. In the Zambezi river basin, priority of delivering available water was given in the following order; 1. Environmental flows, 2. Municipal demands, 3. Mining, 4. Hydropower, 5. Irrigation. In the remaining six river basins, individual demand sites were given specific priorities.

Storage reservoirs and their associated evaporation losses were included in the WEAP modeling. Planned and existing hydropower associated with reservoirs as well as run of the river hydropower facilities were also modeled. In addition, any major diversions

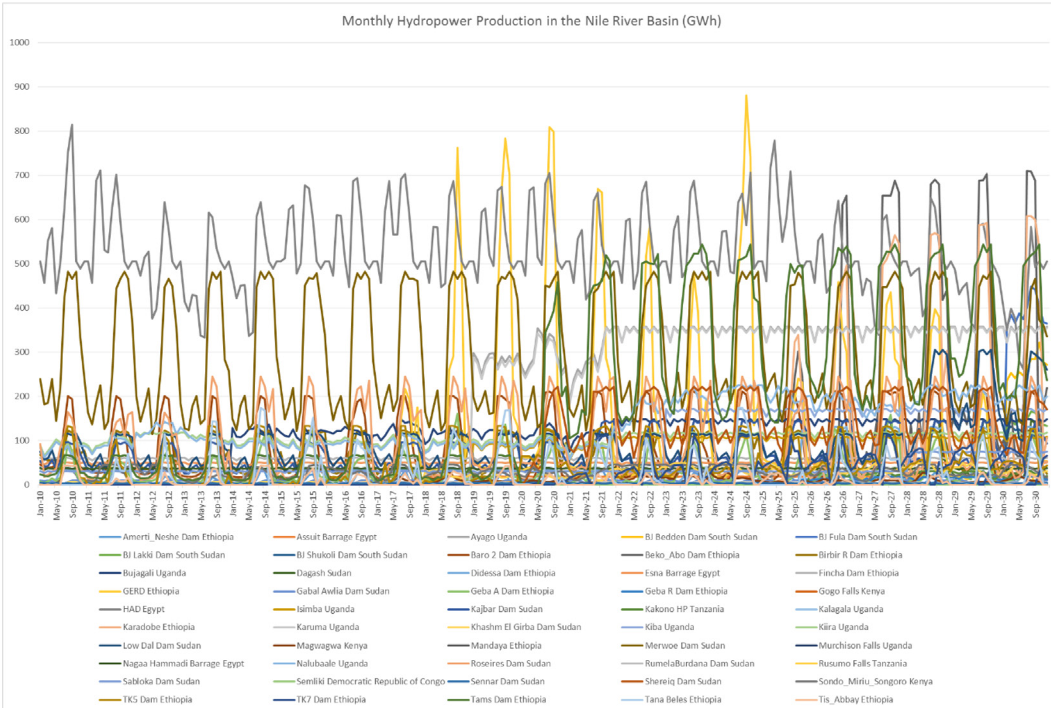

Fig. 12. Monthly Hydropower Production in the Nile River Basin (Gwh) from WEAP analysis.

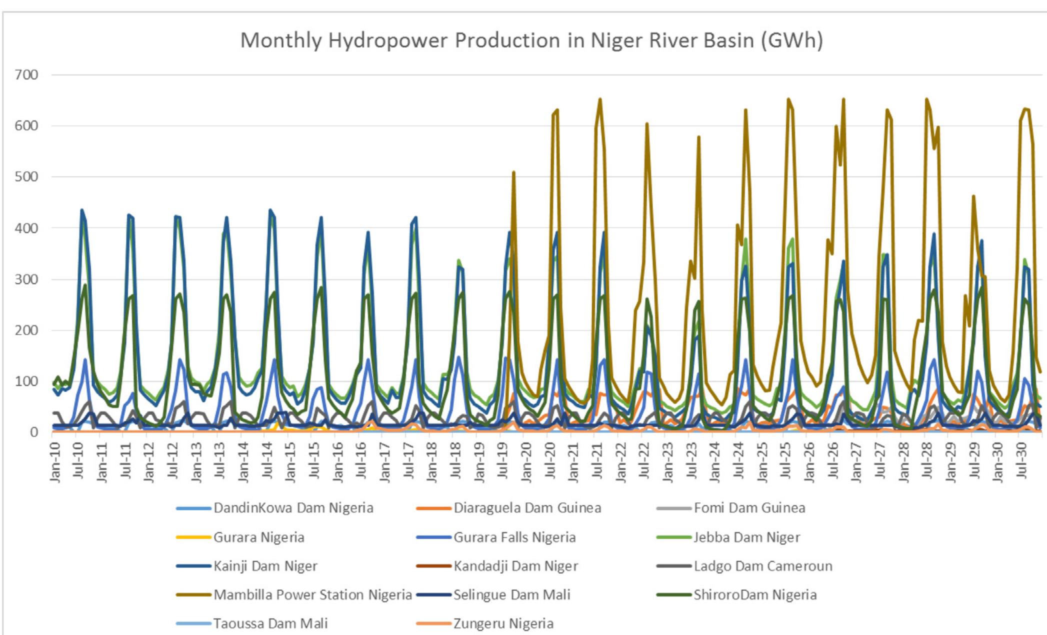

Fig. 13. Monthly Hydropower Production in the Niger River Basin (Gwh) from WEAP analysis.

specific to each river basin were included in the basin representations. The WEAP modeling covered the years from 2010 to 2050 on a monthly time step. Fig. 3–9 show the WEAP river basin representations for the seven river basins modeled.

Methodology

The WEAP river basin models executed the supply-demand balance for each of the seven river basins. The monthly hydropower

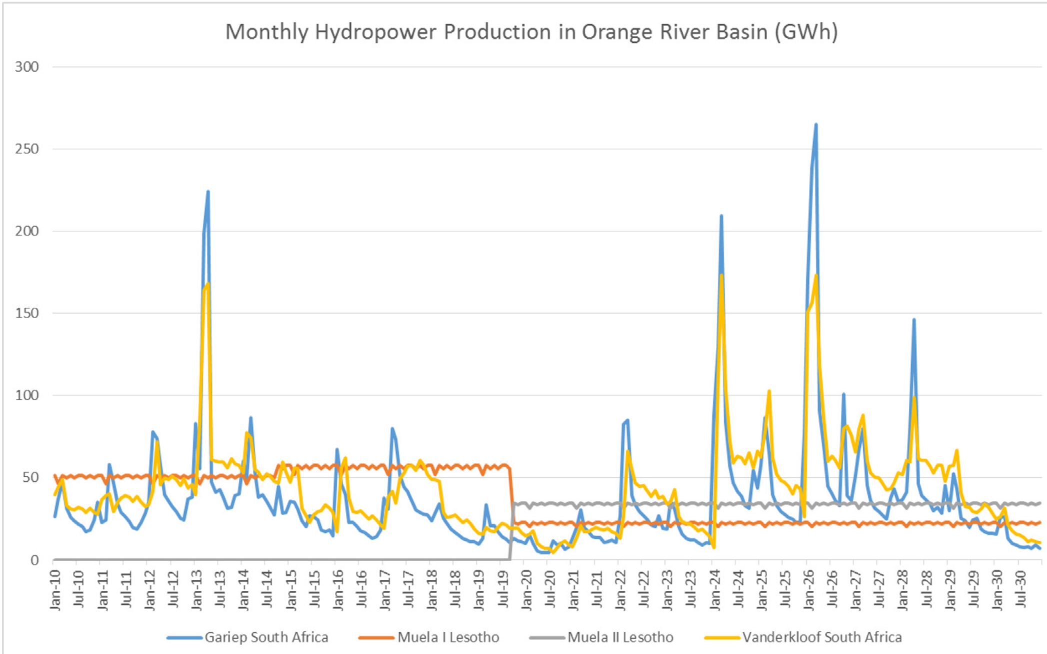

Fig. 14. Monthly Hydropower Production in the Orange River Basin (Gwh) from WEAP analysis.

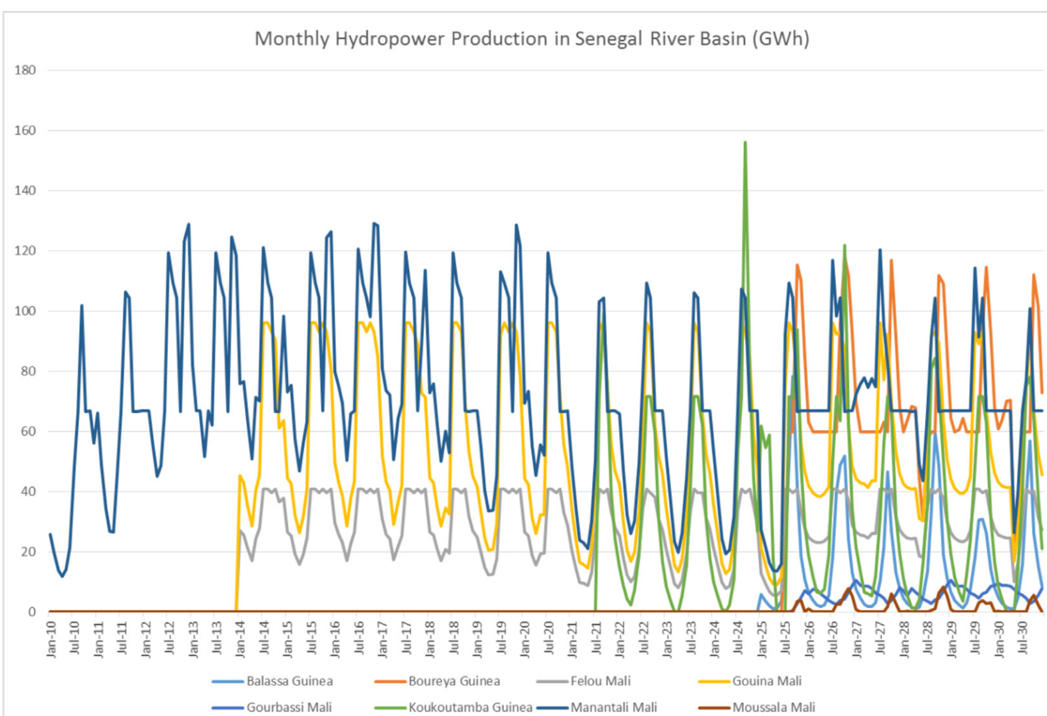

Fig. 15. Monthly Hydropower Production in the Senegal River Basin (Gwh) from WEAP analysis.

production from 2010 to 2050 was produced for each hydropower facility modeled. A list of these facilities can be found in Table A3 in Appendix 3. The hydropower production took into account the other uses in the basins and only provided the amount of hydropower that could be produced given a specific climate scenario and after other higher prioritized demands were met first.

The hydropower facilities were linked to a country. The hydropower production was then summarized into country level pro-

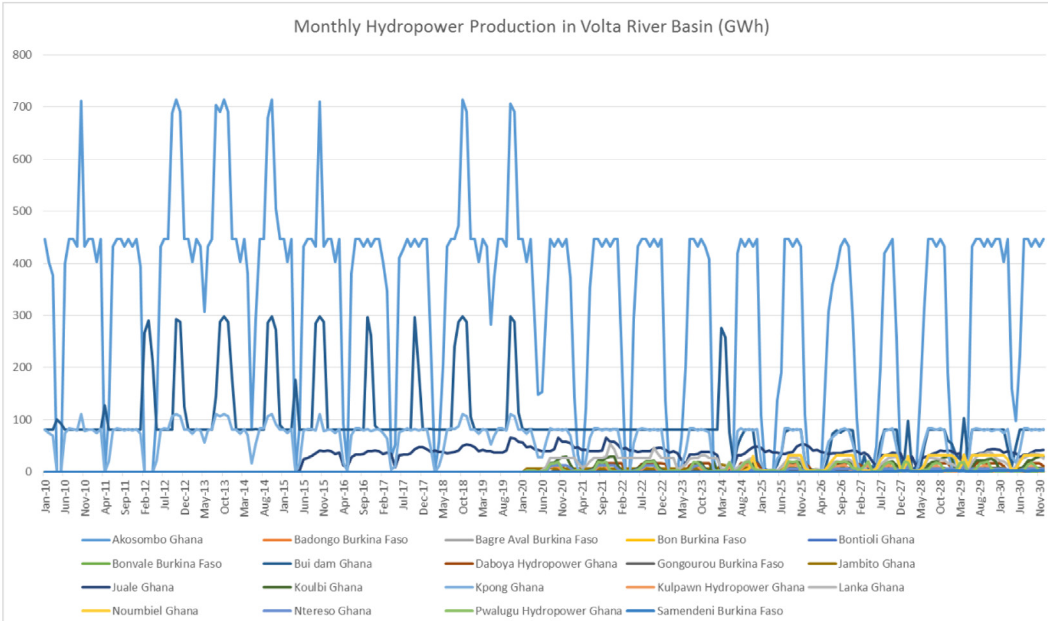

Fig. 16. Monthly Hydropower Production in the Volta River Basin (GWh) from WEAP analysis.

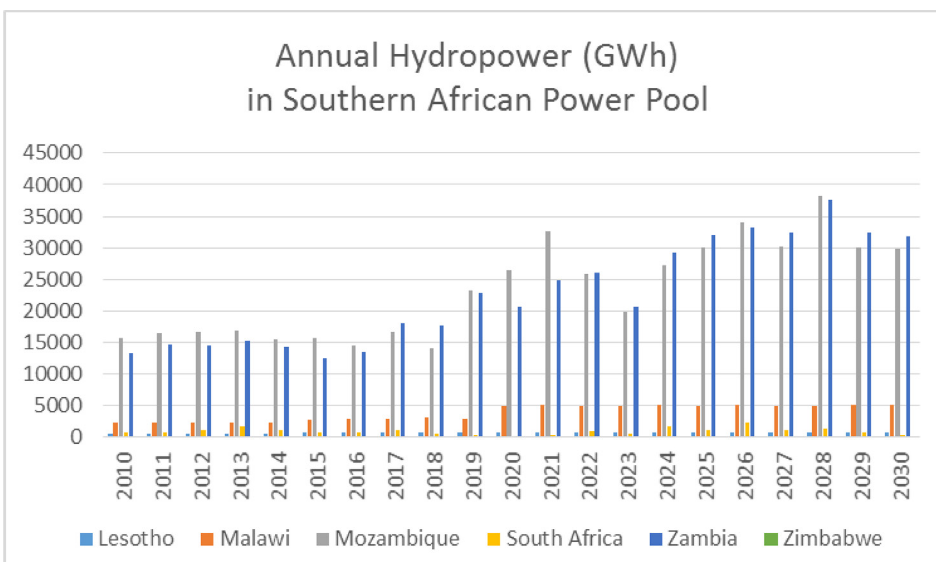

Fig. 17. Annual Hydropower Production in the Southern African Power Pool by Country.

duction values. The countries used in the WEAP modeling were then mapped to the countries used in the RTEP model as shown in Table 2. The country level monthly hydropower from 2010 to 2035 produced by the WEAP model was then read into the RTEP input file for hydropower production. Lastly, the RTEP model was run for the 3 different trade scenarios; No Trade, Min Trade, and Full Trade.

Results

In the following section, the hydropower production produced from the WEAP river basin analyses are shown in Figs. 10–17. The results are from one climate scenario. Following the results of the hydropower production, the results from the three trade scenarios of the RTEP model are presented.

WEAP River Basin Modeling Results

The series of results below show the annual hydropower production by country in Figs. 17–20. The countries were grouped by their power pool association. The results are shown for 2010 through 2030. No hydropower was produced in Zimbabwe over this period, Fig. 17. The planned Batoka Gorge facility was the only hydropower plant linked to Zimbabwe in this study and it did not

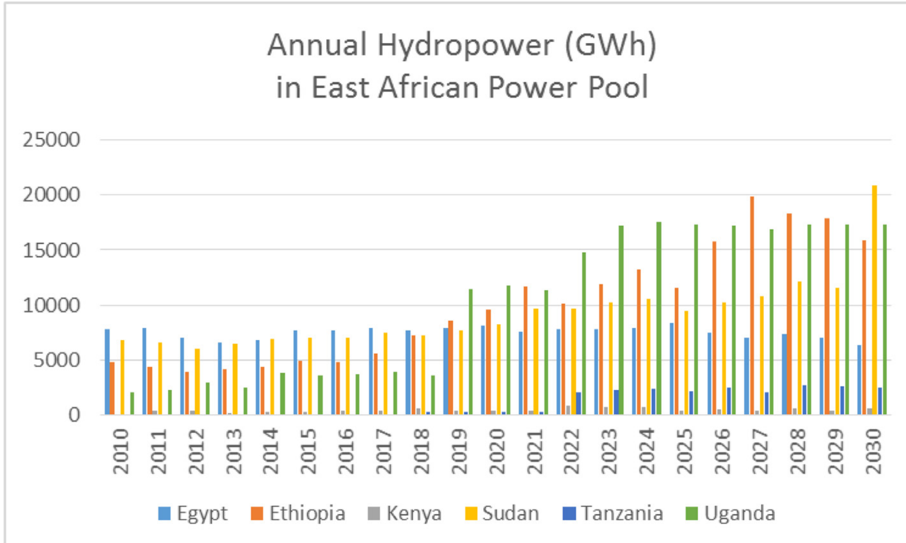

Fig. 18. Annual Hydropower Production in the East African Power Pool by Country.

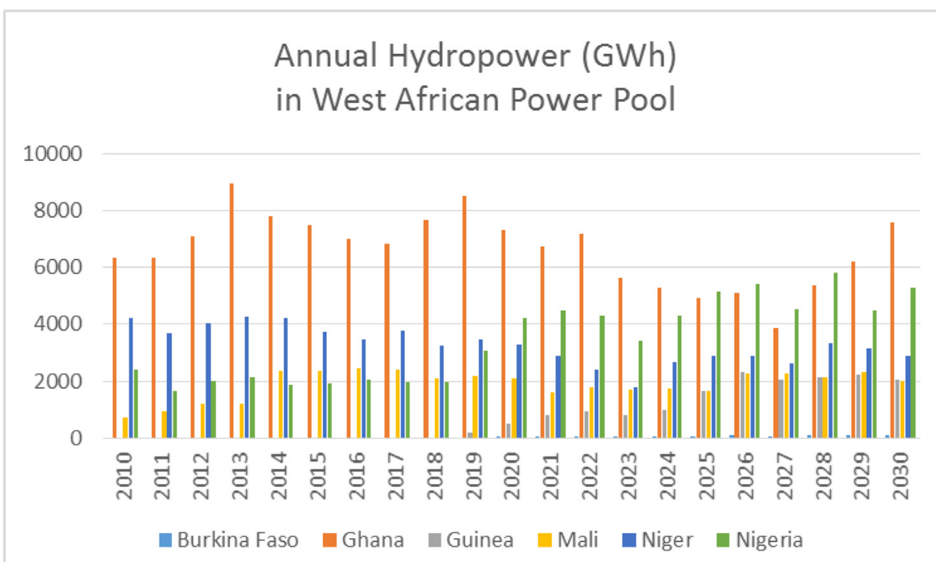

Fig. 19. Annual Hydropower Production in the West African Power Pool by Country.

begin producing hydropower under this climate scenario until after 2030. Also, in Fig. 20, one can see that Cameroon's contribution to hydropower production was extremely small in comparison to Central African Republic and Democratic Republic of Congo.

Fig. 21 shows the annual hydropower production by the four different African power pools. Hydropower production was dominated in the later part of our study period by the Central African Power Pool. It was the Democratic Republic of Congo and more specifically the Inga hydropower projects that produced significantly more hydropower than others starting around 2016 through 2030. The West African Power Pool produced the least amount of hydropower; this became more significant starting in the 2020's when the other regions started new hydropower projects.

RTEP trade scenario results

The RTEP model ran for the 3 different trade scenarios; No Trade, Min Trade, and Full Trade. Figs. 22–24 show the electricity produced by fuel type for the four power pools over the study period 2010 to 2030. Tables 4, 6, and 8 show the sum of the electricity produced over the entire study period by fuel type and power pool. Tables 5, 7, and 9 show the sum of the new capacity for electricity production over the entire study period by fuel type and power pool.

In the Base Case, or No Trade scenario, EAPP produced electricity using mostly gas followed by solar power, then hydropower, geothermal, and distillate. Distillate was a catch-all category con-

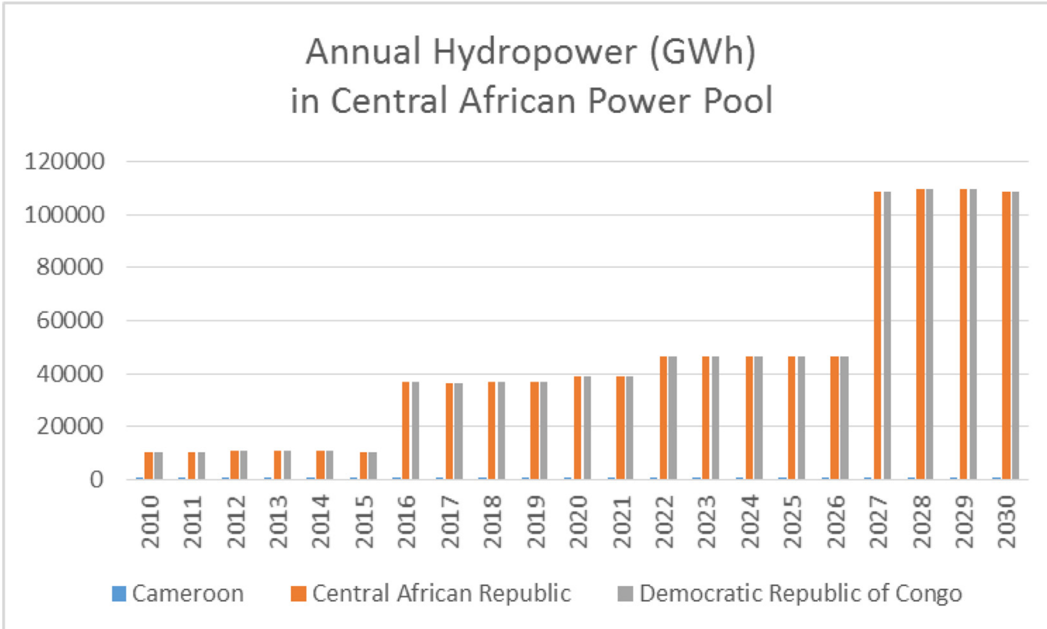

Fig. 20. Annual Hydropower Production in the Central African Power Pool by Country.

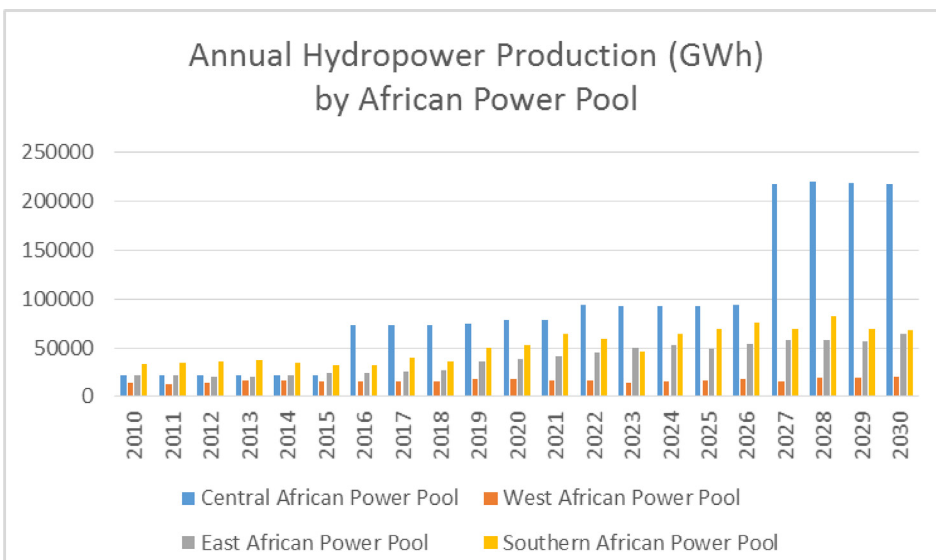

Fig. 21. Annual Hydropower Production by African Power Pool.

taining fuels like diesel and heavy fuel oil. Coal was not used in the EAPP. There was a small amount of hydropower that was not used and spilled. In the same No Trade scenario, SAPP produced electricity using mostly coal and also had unused hydropower. WAPP used mostly gas while CAPP used mostly hydropower to produce electricity in the No Trade scenario. CAPP had a significant amount of unused hydropower (represented by "spill"). These observations were visualized in Fig. 22 and Table 3. When there was no trading among power pools, there was approximately 170 GW of new production capacity installed over the study period over all four power pools (Table 4).

In the Min Trade scenario, electricity production by fuel type was very similar to when there was no trade (Fig. 23 and Table 5). The benefit of allowing even minimum trade was shown in a reduction of new production capacity (Table 6). Overall, across the entire period and all regions, there was a reduction of about 19 GW in new capacity when minimum trade was allowed. The biggest reductions were found in SAPP's coal and distillate new capacity and CAPP's distillate new capacity. By reducing the need to expand production capacity, these regions saved money by not needing to build new facilities but instead were able to use electricity from other power pools.

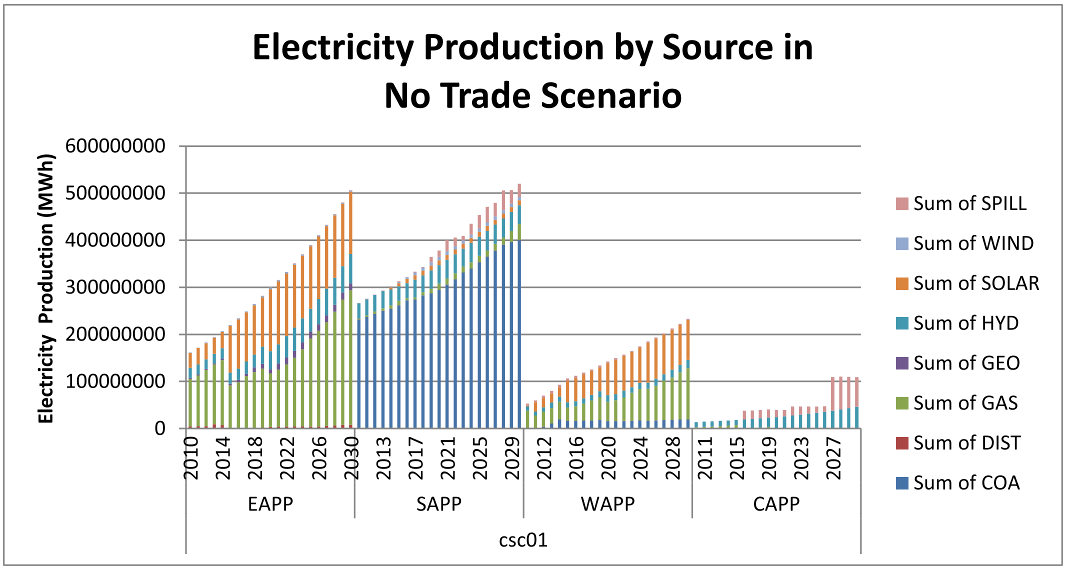

Fig. 22. Electricity Production (MWh) in No Trade RTEP Scenario.

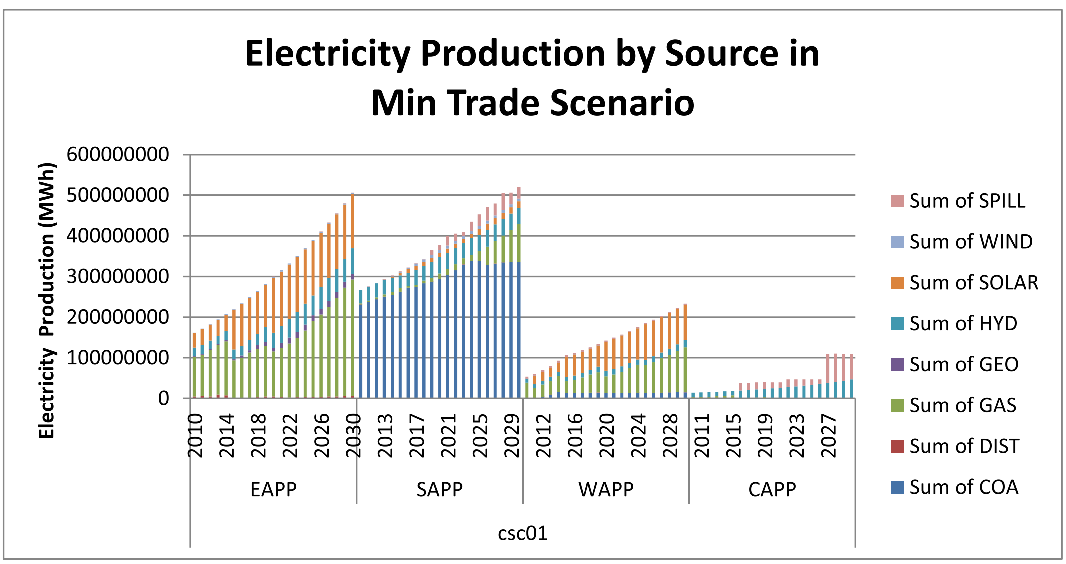

Fig. 23. Electricity Production (MWh) in Min Trade RTEP Scenario.

The regions typically used the same sources for electricity production in the Full Trade scenario as in the No and Min Trade scenarios (Fig. 25 and Table 7). The most notable differences were in the WAPP where they no longer used solar and significantly decreased their use of gas and coal. EAPP increased their use of distillate, gas, and solar. SAPP decreased their use of gas and increased their use of solar. CAPP increased their production of hydropower. In addition, there was no unused hydropower (no spill) in any of the power pools for the Full Trade scenario.

Overall, across the entire period and all regions, there was a reduction of about 10 GW in new capacity compared to the minimum trade scenario (about 30 GW less new capacity than the No Trade scenario). The biggest reductions between the Min and Full Trade scenarios were found in SAPP's coal and gas new capacity (Table 8). WAPP did not add any additional capacity for coal or solar in the Full Trade scenario. WAPP also used significantly less gas for electricity production. CAPP's gas new capacity was less than what was added in the Min Trade scenario. As mentioned previously, by reducing the need to expand production capacity, these regions saved money by not needing to build new facilities but instead were able to use electricity from other power pools.

The amount of electricity traded both within each power pool and between power pools for the three trade scenarios was shown in Table 9. The amount of electricity traded within the SAPP and EAPP increased as external trading increased. In the CAPP, the amount traded within the power pool decreased just slightly between the No Trade and Min Trade scenarios and then increased immensely in the Full Trade scenario. The WAPP decreased the amount of electricity traded internally as one moves from the No Trade to Min Trade and Full Trade scenarios.

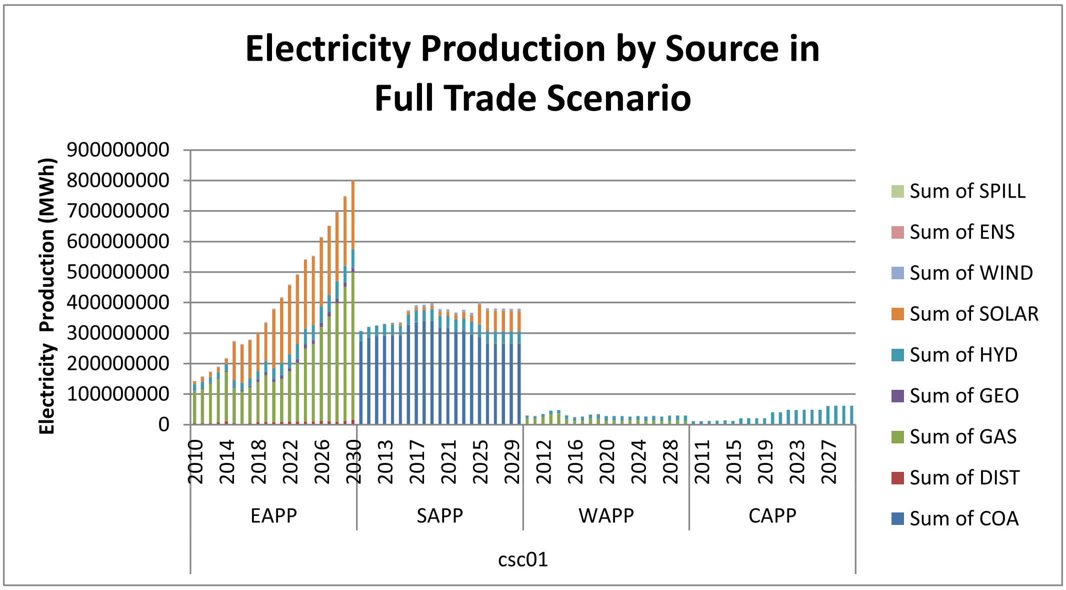

Fig. 24. Electricity Production (MWh) in Full Trade RTEP Scenario.

Table 3
Total electricity production (TWh) over study period (2010-2030) for No Trade Scenario.

	COAL	DISTILLATE	GAS	GEOTHERMAL	HYDRO	SOLAR	WIND	SPILL
EAPP	-	95	3,218	198	801	2,147	46	6
SAPP	6,467	2	236	-	789	165	106	296
WAPP	309	2	1,138	-	265	1,205	0	72
CAPP	10	0	28	-	519	-	-	447

Table 4
Total new electricity production capacity (MW) in the no trade scenario.

	BIO	COA	DIST	GAS	GEO	HYD	NUC	SOLAR	WIND	TOTAL
SAPP	-	22,999	11,072	17,364	-	-	-	-	-	51,434
EAPP	-	-	16,292	29,885	-	-	-	24,564	-	70,741
WAPP	-	2,564	5,663	17,387	-	-	-	16,795	-	42,409
CAPP	-	142	3,154	1,650	-	-	-	-	-	4,947
TOTAL	-	25,705	36,181	66,287	-	-	-	41,360	-	**1,69,532**

Table 5
Total electricity production (TWh) over study period (2010-2030) for min trade scenario.

	COAL	DISTILLATE	GAS	GEOTHERMAL	HYDRO	SOLAR	WIND	SPILL
EAPP	**-**	**86**	**3194**	**198**	**801**	**2178**	**46**	**7**
SAPP	**6178**	**2**	**494**	**-**	**789**	**195**	**106**	**296**
WAPP	**247**	**2**	**1164**	**-**	**264**	**1242**	**0**	**74**
CAPP	**11**	**0**	**26**	**-**	**523**	**-**	**-**	**443**

Table 6
Total new electricity production capacity (MW) in the min trade scenario.

	BIO	COA	DIST	GAS	GEO	HYD	NUC	SOLAR	WIND	
SAPP	-	12,059	2,311	27,281	-	-	-	922	-	42,572
EAPP	-	-	10,466	30,079	-	-	-	24,800	-	65,345
WAPP	-	2,034	4,310	17,543	-	-	-	17,253	-	41,140
CAPP	-	180	89	1,610	-	-	-	-	-	1,879
TOTAL	-	14,272	17,176	76,513	-	-	-	42,975	-	**1,50,936**

Table 7
Total electricity production (TWh) over study period (2010-2030) for full trade scenario.

	COAL	DISTILLATE	GAS	GEOTHERMAL	HYDRO	SOLAR	WIND	SPILL
EAPP	-	184	4,357	185	801	3,131	46	-
SAPP	6,219	1	22	-	789	543	106	-
WAPP	4	0	355	-	272	0	0	-
CAPP	15	0	20	-	696	-	-	-

Table 8
Total new electricity production capacity (MW) in the full trade scenario.

	BIO	COA	DIST	GAS	GEO	HYD	NUC	SOLAR	WIND	
SAPP	**-**	2584	2954	1833	0	0	0	10311	0	17,682
EAPP	**-**	0	10909	65514	0	0	0	41857	0	1,18,279
WAPP	**-**	0	3670	174	0	0	0	0	0	3,844
CAPP	**-**	109	551	561	0	0	0	0	0	1,221
TOTAL	-	2,693	18,085	68,081	-	-	-	52,167	-	**1,41,026**

Table 9
Electricity Traded both within and between different power pools.

	No Trade (MWh)	Min Trade (MWh)	Full Trade (MWh)
SAPP-SAPP	24,55,988	28,30,052	45,07,738
SAPP-EAPP		121	1,78,945
SAPP-WAPP		-	-
SAPP-CAPP		19	7,11,417
EAPP-SAPP		108	4,29,633
EAPP-EAPP	15,10,250	16,10,388	47,19,930
EAPP-WAPP		-	-
EAPP-CAPP		47	20,78,328
WAPP-SAPP		-	-
WAPP-EAPP		-	-
WAPP-WAPPP	39,62,632	37,77,760	32,31,147
WAPP-CAPP		-	35,732
CAPP-SAPP		2,134	5,49,240
CAPP-EAPP		2,085	73,876
CAPP-WAPP		-	23,81,123
CAPP-CAPP	7,63,890	7,47,167	79,02,539

Table A1.1
SAPP Country level electricity load levels for each month and time slice for the year 2010 (GW) [5].

		Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
ANGOLA	L1	0.84	0.83	0.83	0.83	0.82	0.82	0.81	0.82	0.82	0.83	0.83	0.83
	L2	0.77	0.77	0.77	0.77	0.76	0.76	0.76	0.76	0.76	0.77	0.77	0.77
	L3	0.69	0.68	0.68	0.67	0.67	0.66	0.65	0.66	0.67	0.67	0.68	0.68
BOTSWANA	L1	0.49	0.49	0.49	0.48	0.48	0.48	0.48	0.48	0.48	0.48	0.49	0.49
	L2	0.48	0.48	0.47	0.47	0.47	0.47	0.46	0.47	0.47	0.47	0.47	0.48
	L3	0.46	0.46	0.45	0.45	0.44	0.44	0.44	0.44	0.44	0.45	0.45	0.46
LESOTHO	L1	0.11	0.11	0.11	0.11	0.11	0.11	0.11	0.11	0.11	0.11	0.11	0.11
	L2	0.06	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05
	L3	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05
MALAWI	L1	0.24	0.24	0.23	0.23	0.23	0.23	0.23	0.23	0.23	0.23	0.23	0.24
	L2	0.22	0.22	0.22	0.22	0.22	0.22	0.21	0.22	0.22	0.22	0.22	0.22
	L3	0.21	0.21	0.21	0.21	0.21	0.21	0.21	0.21	0.21	0.21	0.21	0.21
MOZAMBIQUE	L1	0.60	0.60	0.59	0.59	0.59	0.59	0.58	0.59	0.59	0.59	0.59	0.60
	L2	0.44	0.44	0.43	0.43	0.43	0.43	0.43	0.43	0.43	0.43	0.43	0.44
	L3	0.39	0.39	0.39	0.39	0.38	0.38	0.38	0.38	0.38	0.39	0.39	0.39
NAMIBIA	L1	0.56	0.56	0.56	0.56	0.55	0.55	0.55	0.55	0.55	0.56	0.56	0.56
	L2	0.45	0.45	0.45	0.45	0.44	0.44	0.44	0.44	0.44	0.45	0.45	0.45
	L3	0.40	0.39	0.39	0.39	0.39	0.39	0.39	0.39	0.39	0.39	0.39	0.39
SOUTH AFRICA	L1	31.78	31.60	31.43	31.34	31.16	30.99	30.90	30.99	31.16	31.34	31.43	31.60
	L2	26.56	26.38	26.20	26.12	25.94	25.76	25.67	25.76	25.94	26.12	26.20	26.38
	L3	22.75	22.57	22.40	22.22	22.04	21.87	21.69	21.87	22.04	22.22	22.40	22.57
SWAZILAND	L1	0.18	0.18	0.18	0.18	0.18	0.18	0.18	0.18	0.18	0.18	0.18	0.18
	L2	0.15	0.15	0.15	0.15	0.15	0.15	0.15	0.15	0.15	0.15	0.15	0.15
	L3	0.13	0.13	0.13	0.13	0.13	0.13	0.13	0.13	0.13	0.13	0.13	0.13
ZAMBIA	L1	1.26	1.25	1.25	1.24	1.23	1.23	1.22	1.23	1.23	1.24	1.25	1.25
	L2	1.12	1.12	1.11	1.11	1.10	1.10	1.09	1.10	1.10	1.11	1.11	1.12
	L3	1.05	1.04	1.04	1.03	1.03	1.02	1.02	1.02	1.03	1.03	1.04	1.04
ZIMBABWE	L1	1.37	1.36	1.36	1.35	1.34	1.34	1.33	1.34	1.34	1.35	1.36	1.36
	L2	1.14	1.14	1.13	1.13	1.12	1.12	1.11	1.12	1.12	1.13	1.13	1.14
	L3	0.98	0.98	0.97	0.97	0.96	0.96	0.95	0.96	0.96	0.97	0.97	0.98

Table A1.2
EAPP Country load levels for each month and time slice for the year 2010 (GW) [5].

		Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
BURUNDI	L1	0.03	0.03	0.03	0.03	0.03	0.03	0.03	0.03	0.03	0.03	0.03	0.03
	L2	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01
	L3	0.008	0.008	0.008	0.008	0.008	0.008	0.008	0.008	0.008	0.008	0.008	0.008
DJIBOUTI	L1	0.018	0.019	0.020	0.021	0.023	0.025	0.027	0.018	0.019	0.020	0.021	0.023
	L2	0.020	0.021	0.022	0.023	0.025	0.028	0.030	0.020	0.021	0.022	0.023	0.025
	L3	0.020	0.021	0.022	0.024	0.025	0.028	0.030	0.020	0.021	0.022	0.024	0.025
EGYPT	L1	18.33	19.09	19.92	20.83	21.82	22.91	22.91	21.82	20.83	19.92	19.09	18.33
	L2	13.75	14.32	14.94	15.62	16.37	17.18	17.18	16.37	15.62	14.94	14.32	13.75
	L3	11.00	11.46	11.95	12.50	13.09	13.75	13.75	13.09	12.50	11.95	11.46	11.00
ETHIOPIA	L1	0.88	0.73	0.73	0.76	0.73	0.73	0.70	0.73	0.85	0.82	0.76	0.82
	L2	0.78	0.65	0.65	0.68	0.65	0.65	0.63	0.65	0.76	0.73	0.68	0.73
	L3	0.54	0.45	0.45	0.46	0.45	0.45	0.43	0.45	0.52	0.50	0.46	0.50
KENYA	L1	1.08	1.08	1.08	1.08	1.08	1.08	1.08	1.08	1.08	1.08	1.08	1.08
	L2	0.72	0.72	0.72	0.72	0.72	0.72	0.72	0.72	0.72	0.72	0.72	0.72
	L3	0.64	0.64	0.64	0.64	0.64	0.64	0.64	0.64	0.64	0.64	0.64	0.64
RWANDA	L1	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05
	L2	0.03	0.03	0.03	0.03	0.03	0.03	0.03	0.03	0.03	0.03	0.03	0.03
	L3	0.02	0.02	0.02	0.02	0.02	0.02	0.02	0.02	0.02	0.02	0.02	0.02
SUDAN	L1	0.79	0.66	0.66	0.68	0.66	0.66	0.63	0.66	0.76	0.73	0.68	0.73
	L2	1.36	1.13	1.13	1.18	1.13	1.13	1.09	1.13	1.31	1.27	1.18	1.27
	L3	0.53	0.44	0.44	0.46	0.44	0.44	0.42	0.44	0.51	0.49	0.46	0.49
TANZANIA	L1	0.77	0.77	0.77	0.77	0.77	0.77	0.77	0.77	0.77	0.77	0.77	0.77
	L2	0.51	0.51	0.51	0.51	0.51	0.51	0.51	0.51	0.51	0.51	0.51	0.51
	L3	0.45	0.45	0.45	0.45	0.45	0.45	0.45	0.45	0.45	0.45	0.45	0.45
UGANDA	L1	0.60	0.60	0.60	0.60	0.60	0.60	0.60	0.60	0.60	0.60	0.60	0.60
	L2	0.39	0.39	0.39	0.39	0.39	0.39	0.39	0.39	0.39	0.39	0.39	0.39
	L3	0.28	0.28	0.28	0.28	0.28	0.28	0.28	0.28	0.28	0.28	0.28	0.28

Table A1.3
WAPP Country load levels for each month and time slice for the year 2010 (GW) [5].

		Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
BURKINA FASO	L1	0.09	0.10	0.13	0.16	0.18	0.16	0.15	0.09	0.15	0.15	0.12	0.11
	L2	0.07	0.08	0.10	0.12	0.13	0.12	0.11	0.07	0.11	0.12	0.10	0.08
	L3	0.07	0.08	0.10	0.12	0.13	0.12	0.11	0.07	0.11	0.11	0.09	0.08
COTE D'IVOIRE	L1	0.87	0.81	0.90	0.87	0.88	0.84	0.86	0.86	0.85	0.90	0.90	0.94
	L2	0.73	0.68	0.75	0.73	0.74	0.70	0.71	0.71	0.61	0.65	0.65	0.68
	L3	0.61	0.57	0.63	0.61	0.62	0.59	0.60	0.60	0.52	0.55	0.55	0.57
THE GAMBIA	L1	0.04	0.04	0.04	0.03	0.03	0.03	0.04	0.04	0.04	0.03	0.04	0.05
	L2	0.02	0.03	0.03	0.02	0.02	0.02	0.03	0.03	0.03	0.03	0.03	0.03
	L3	0.02	0.03	0.03	0.02	0.03	0.02	0.03	0.03	0.03	0.02	0.02	0.03
GHANA	L1	1.23	1.14	1.27	1.23	1.25	1.18	1.21	1.21	1.19	1.26	1.26	1.32
	L2	1.02	0.95	1.06	1.02	1.04	0.98	1.01	1.01	0.86	0.91	0.91	0.96
	L3	0.86	0.80	0.89	0.86	0.87	0.83	0.84	0.84	0.73	0.77	0.77	0.81
GUINEA	L1	0.09	0.10	0.10	0.10	0.11	0.11	0.13	0.12	0.10	0.14	0.11	0.10
	L2	0.07	0.07	0.08	0.07	0.08	0.08	0.10	0.09	0.07	0.10	0.08	0.08
	L3	0.05	0.06	0.06	0.06	0.07	0.07	0.08	0.07	0.06	0.09	0.07	0.06
GUINEA-BISSAU	L1	0.02	0.02	0.02	0.02	0.02	0.02	0.03	0.02	0.02	0.03	0.02	0.02
	L2	0.01	0.01	0.02	0.02	0.02	0.02	0.02	0.02	0.02	0.02	0.02	0.02
	L3	0.01	0.01	0.01	0.01	0.01	0.01	0.02	0.02	0.01	0.02	0.01	0.01
LIBERIA	L1	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01
	L2	0.01	0.00	0.01	0.01	0.01	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	L3	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
MALI	L1	0.10	0.11	0.14	0.17	0.19	0.17	0.16	0.10	0.16	0.16	0.13	0.12
	L2	0.08	0.09	0.11	0.14	0.15	0.14	0.13	0.08	0.13	0.13	0.11	0.10
	L3	0.08	0.09	0.11	0.14	0.15	0.14	0.13	0.08	0.13	0.14	0.11	0.10
NIGER	L1	0.08	0.09	0.11	0.13	0.15	0.13	0.12	0.08	0.12	0.13	0.10	0.09
	L2	0.07	0.07	0.09	0.11	0.12	0.11	0.10	0.06	0.10	0.11	0.09	0.08
	L3	0.06	0.07	0.09	0.11	0.12	0.11	0.10	0.06	0.10	0.11	0.09	0.08
NIGERIA	L1	3.69	3.42	3.47	3.20	3.31	3.26	2.39	2.22	2.17	2.17	2.33	2.60
	L2	3.78	3.50	3.56	3.28	3.39	3.34	2.45	2.28	2.22	2.22	2.39	2.67
	L3	3.65	3.38	3.43	3.16	3.27	3.22	2.36	2.20	2.14	2.14	2.30	2.57
SENEGAL	L1	0.30	0.33	0.35	0.33	0.37	0.38	0.43	0.39	0.34	0.43	0.37	0.35
	L2	0.26	0.29	0.30	0.29	0.32	0.33	0.38	0.34	0.29	0.38	0.33	0.31
	L3	0.22	0.24	0.26	0.25	0.28	0.29	0.32	0.29	0.25	0.32	0.28	0.26
SIERRA LEONE	L1	0.02	0.02	0.02	0.02	0.02	0.02	0.03	0.02	0.02	0.03	0.02	0.02
	L2	0.02	0.02	0.02	0.02	0.02	0.02	0.02	0.02	0.02	0.02	0.02	0.02
	L3	0.01	0.01	0.02	0.01	0.02	0.02	0.02	0.02	0.01	0.02	0.02	0.02
TOGO/BENIN	L1	0.32	0.29	0.33	0.32	0.32	0.30	0.31	0.31	0.31	0.33	0.32	0.34
	L2	0.26	0.24	0.27	0.26	0.27	0.25	0.26	0.26	0.22	0.23	0.23	0.25
	L3	0.22	0.21	0.23	0.22	0.22	0.21	0.22	0.22	0.19	0.20	0.20	0.21

Table A1.4
CAPP Country load levels for each month and time slice for the year 2010 (GW) [5].

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
CAMEROON	0.71	0.70	0.70	0.69	0.69	0.69	0.67	0.65	0.68	0.70	0.72	0.74
	0.51	0.50	0.50	0.50	0.50	0.50	0.48	0.47	0.49	0.50	0.52	0.53
	0.47	0.47	0.46	0.46	0.46	0.46	0.44	0.43	0.45	0.47	0.48	0.49
CENTRAL AFRICAN REPUBLIC (CAR)	0.03	0.03	0.03	0.03	0.03	0.03	0.03	0.03	0.03	0.03	0.03	0.03
	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01
	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01
CHAD	0.02	0.02	0.02	0.03	0.03	0.03	0.03	0.02	0.03	0.03	0.02	0.02
	0.01	0.01	0.02	0.02	0.02	0.02	0.02	0.01	0.02	0.02	0.02	0.02
	0.01	0.01	0.02	0.02	0.02	0.02	0.02	0.01	0.02	0.02	0.02	0.02
CONGO	0.10	0.10	0.09	0.09	0.09	0.08	0.08	0.08	0.09	0.09	0.09	0.10
	0.07	0.07	0.06	0.06	0.06	0.06	0.06	0.06	0.06	0.06	0.06	0.07
	0.07	0.07	0.07	0.07	0.07	0.07	0.06	0.07	0.07	0.07	0.07	0.07
DEMOCRATIC REPUBLIC OF CONGO (DRC)	1	0.95	0.91	0.86	0.81	0.77	0.73	0.77	0.82	0.86	0.91	0.95
	0.85	0.81	0.77	0.75	0.71	0.67	0.63	0.67	0.71	0.73	0.77	0.81
	0.94	0.89	0.85	0.85	0.81	0.76	0.72	0.76	0.81	0.81	0.85	0.89
EQUATORIAL GUINEA	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01
	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01
	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01
GABON	0.175	0.17	0.17	0.17	0.17	0.17	0.17	0.17	0.17	0.17	0.17	0.17
	0.15	0.15	0.15	0.15	0.15	0.14	0.14	0.14	0.15	0.14	0.15	0.15
	0.16	0.16	0.16	0.1	0.17	0.16	0.16	0.16	0.17	0.16	0.16	0.16

Table A2.1
SAPP electricity demand growth assumptions [5].

Country	2010-2014	2015-2019	2020-2024	2025-2030
ANGOLA	6%	6%	5%	5%
BOTSWANA	4%	4%	4%	4%
LESOTHO	4%	4%	4%	4%
MALAWI	4%	4%	3%	3%
MOZAMBIQUE	4%	4%	4%	4%
NAMIBIA	2%	2%	1%	1%
SOUTH AFRICA	3%	3%	3%	3%
SWAZILAND	3%	3%	1%	1%
ZAMBIA	3%	3%	3%	3%
ZIMBABWE	5%	5%	4%	4%

Table A2.2
EAPP electricity demand growth assumptions [5].

Country	2010-2014	2015-2019	2020-2024	2025-2030
BURUNDI	14%	9%	6%	5%
DJIBOUTI	5%	5%	5%	5%
EGYPT	6%	6%	5%	5%
ETHIOPIA	8%	8%	7%	7%
KENYA	6%	6%	5%	5%
RWANDA	6%	6%	5%	4%
SUDAN	12%	10%	9%	8%
TANZANIA	7%	6%	5%	5%
UGANDA	5%	5%	6%	5%

Table A2.3
WAPP electricity demand growth assumptions [5].

Country	2010-2014	2015-2019	2020-2024	2025-2030
BURKINA FASO	6%	7%	7%	7%
COTE D'IVOIRE	5%	6%	5%	5%
THE GAMBIA	18%	9%	3%	4%
GHANA	10%	5%	5%	5%
GUINEA	13%	26%	2%	2%
GUINEA-BISSAU	4%	19%	10%	2%
LIBERIA	53%	15%	1%	1%
MALI	14%	8%	5%	4%
NIGER	5%	7%	5%	4%
NIGERIA	21%	6%	6%	5%
SENEGAL	7%	8%	5%	5%
SIERRA LEONE	37%	22%	4%	1%
TOGO/BENIN	10%	8%	7%	6%

Table A2.4
CAPP electricity demand growth assumptions [5].

Country	2010-2014	2015-2019	2020-2024	2025-2030
CAMEROON	9%	9%	9%	9%
CAR	8%	8%	7%	7%
CHAD	7%	7%	6%	6%
CONGO	4%	4%	4%	4%
DRC	4%	4%	4%	4%
EQUATORIAL GUINEA	15%	10%	10%	8%
GABON	4%	4%	4%	4%

Table A3.1
Planned or existing hydropower facilities modeled in WEAP.

WEAP Area	Hydro Facility	Country	Planned or Existing (Annual Energy Demand GWh)
Congo	Busanga	Democratic Republic of Congo	Planned (Not Set)
Congo	Grand Inga	Democratic Republic of Congo	Planned (250000)
Congo	Inga 1	Democratic Republic of Congo	Existing (2400)
Congo	Inga 2	Democratic Republic of Congo	Existing (Not Set)
Congo	Inga 3	Democratic Republic of Congo	Planned (Not Set)
Congo	Katende	Democratic Republic of Congo	Planned (Not Set)
Congo	Mobayi	Democratic Republic of Congo	Planned (Not Set)
Congo	Nseke	Democratic Republic of Congo	Planned (Not Set)
Congo	Nzilo	Democratic Republic of Congo	Planned (Not Set)
Congo	Palambo Dam	Central African Republic	Planned (Not Set)
Congo	Ruzizi II	Democratic Republic of Congo	Existing (Not Set)
Congo	Ruzizi III	Democratic Republic of Congo	Planned (Not Set)
Congo	Sanga	Democratic Republic of Congo	Planned (Not Set)
Congo	Tshopo	Democratic Republic of Congo	Planned (Not Set)
Congo	Zongo	Democratic Republic of Congo	Existing (Not Set)
Zambezi	Batoka Gorge	Zimbabwe	Planned (1618)
Zambezi	Cahora Bassa	Mozambique	Existing (7420)
Zambezi	Devils Gorge	Zambia	Planned (5600)
Zambezi	Iztezhi Tezhi	Zambia	Planned (258)
Zambezi	Kafue Gorge L	Zambia	Planned (2168)
Zambezi	Kafue Gorge U	Zambia	Existing (4292)
Zambezi	Kapichira	Malawi	Existing (394)
Zambezi	Kholombizo	Malawi	Planned (318)
Zambezi	Lake Kariba	Zambia	Planned (5624)
Zambezi	Lower Fufu	Malawi	Planned (134)
Zambezi	Lusiwasi	Zambia	Planned (540)
Zambezi	Lusumfwe_Mulungushi	Zambia	Planned (193)
Zambezi	Mpata Gorge	Zambia	Planned (4200)
Zambezi	Mphanda Nkuwa	Mozambique	Planned (3867)
Zambezi	Nkhula Falls	Malawi	Existing (440)
Zambezi	Rumakali	Tanzania	Planned (670)
Zambezi	Songwe I	Malawi	Planned (29)
Zambezi	Songwe II	Malawi	Planned (228)
Zambezi	Songwe III	Malawi	Planned (197)
Zambezi	Tedzani	Malawi	Existing (281)
Zambezi	Victoria Falls	Zambia	Existing (301)
Nile	Amerti_Neshe Dam	Ethiopia	Existing (467)
Nile	Assuit Barrage	Egypt	Planned (Not Set)
Nile	Ayago	Uganda	Planned (4218)
Nile	BJ Bedden Dam	South Sudan	Planned (3901)
Nile	BJ Fula Dam	South Sudan	Planned (6876)
Nile	BJ Lakki Dam	South Sudan	Planned (2794)
Nile	BJ Shukoli Dam	South Sudan	Planned (1598)
Nile	Baro 2 Dam	Ethiopia	Planned (2496)
Nile	Beko_Abo Dam	Ethiopia	Planned (Not Set)
Nile	Birbir R Dam	Ethiopia	Planned (2444)
Nile	Bujagali	Uganda	Existing (1744)
Nile	Dagash	Sudan	Planned (Not Set)
Nile	Didessa Dam	Ethiopia	Planned (1550)
Nile	Esna Barrage	Egypt	Existing (Not Set)
Nile	Fincha Dam	Ethiopia	Existing (728)
Nile	GERD	Ethiopia	Planned (18396)
Nile	Gabal Awlia Dam	Sudan	Existing (126)
Nile	Geba A Dam	Ethiopia	Planned (5347)
Nile	Geba R Dam	Ethiopia	Planned (226)
Nile	Gogo Falls	Kenya	Existing (14)
Nile	HAD	Egypt	Existing (5960)
Nile	Isimba	Uganda	Planned (1289)
Nile	Kajbar Dam	Sudan	Planned (1702)

(continued on next page)

Table A3.1 (*continued*)

WEAP Area	Hydro Facility	Country	Planned or Existing (Annual Energy Demand GWh)
Nile	Kakono HP	Tanzania	Planned (371)
Nile	Kalagala	Uganda	Planned (2235)
Nile	Karadobe	Ethiopia	Planned (8830)
Nile	Karuma	Uganda	Planned (4190)
Nile	Khashm El Girba Dam	Sudan	Existing (79)
Nile	Kiba	Uganda	Planned (2018)
Nile	Kiira	Uganda	Existing (1394)
Nile	Low Dal Dam	Sudan	Planned (3584)
Nile	Magwagwa	Kenya	Planned (525)
Nile	Mandaya	Ethiopia	Planned (11177)
Nile	Merwoe Dam	Sudan	Existing (5694)
Nile	Murchison Falls	Uganda	Planned (4295)
Nile	Nagaa Hammadi Barrage	Egypt	Existing (Not Set)
Nile	Nalubaale	Uganda	Existing (2663)
Nile	Roseires Dam	Sudan	Existing (2871)
Nile	RumelaBurdana Dam	Sudan	Planned (Not Set)
Nile	Rusumo Falls	Tanzania	Planned (588)
Nile	Sabloka Dam	Sudan	Planned (893)
Nile	Semliki	Democratic Republic of Congo	Planned (490)
Nile	Sennar Dam	Sudan	Existing (175)
Nile	Shereiq Dam	Sudan	Planned (2612)
Nile	Sondo_Miriu_Songoro	Kenya	Existing (546)
Nile	TK5 Dam	Ethiopia	Existing (Not Set)
Nile	TK7 Dam	Ethiopia	Planned (Not Set)
Nile	Tams Dam	Ethiopia	Planned (6407)
Nile	Tana Beles	Ethiopia	Existing (2055)
Nile	Tis_Abbay	Ethiopia	Existing (514)
Niger	DandinKowa Dam	Nigeria	Existing (108)
Niger	Diaraguela Dam	Guinea	Planned (348)
Niger	Fomi Dam	Guinea	Planned (432)
Niger	Gurara	Nigeria	Planned (216)
Niger	Gurara Falls	Nigeria	Planned (Not Set)
Niger	Jebba Dam	Niger	Existing (2256)
Niger	Kainji Dam	Niger	Existing (2712)
Niger	Kandadji Dam	Niger	Planned (624)
Niger	Ladgo Dam	Cameroun	Existing (264)
Niger	Mambilla Power Station	Nigeria	Planned (7680)
Niger	Selingue Dam	Mali	Existing (156)
Niger	ShiroroDam	Nigeria	Existing (1848)
Niger	Taoussa Dam	Mali	Planned (96)
Niger	Zungeru	Nigeria	Planned (4728)
Orange	Gariep	South Africa	Existing (Not Set)
Orange	Muela I	Lesotho	Existing (Not Set)
Orange	Muela II	Lesotho	Planned (Not Set)
Orange	Vanderkloof	South Africa	Existing (Not Set)
Senegal	Balassa	Guinea	Planned (470)
Senegal	Boureya	Guinea	Planned (717)
Senegal	Felou	Mali	Planned (320)
Senegal	Gouina	Mali	Planned (510)
Senegal	Gourbassi	Mali	Planned (104)
Senegal	Koukoutamba	Guinea	Planned (858)
Senegal	Manantali	Mali	Existing (800)
Senegal	Moussala	Mali	Planned (175)
Volta	Akosombo	Ghana	Existing (5256)
Volta	Badongo	Burkina Faso	Planned (24)
Volta	Bagre Aval	Burkina Faso	Planned (110)
Volta	Bon	Burkina Faso	Planned (61)
Volta	Bontioli	Ghana	Planned (40)
Volta	Bonvale	Burkina Faso	Planned (2)
Volta	Bui dam	Ghana	Existing (972)
Volta	Daboya Hydropower	Ghana	Planned (194)
Volta	Gongourou	Burkina Faso	Planned (39)
Volta	Jambito	Ghana	Planned (180)
Volta	Juale	Ghana	Existing (684)
Volta	Koulbi	Ghana	Planned (392)
Volta	Kpong	Ghana	Existing (Not Set)
Volta	Kulpawn Hydropower	Ghana	Planned (165)
Volta	Lanka	Ghana	Planned (324)
Volta	Noumbiel	Ghana	Planned (378)
Volta	Ntereso	Ghana	Planned (256)
Volta	Pwalugu Hydropower	Ghana	Planned (183)
Volta	Samendeni	Burkina Faso	Planned (19)

In the Min Trade scenario, the CAPP exported the most electricity to EAPP and SAPP. The WAPP did not export any electricity while the SAPP and EAPP exported small amounts to other power pools. In the Full Trade scenario, CAPP again exported the most electricity to EAPP and SAPP. The WAPP now exported electricity to the CAPP in the Full Trade scenario. The EAPP exported a large amount of electricity to the CAPP in the Full Trade scenario while the SAPP exported some electricity to the EAPP and CAPP.

The CAPP was the largest importer and exporter because it served as the connection for all other inter-regional trade. The CAPP was the only region that was connected to all other regions. It was the only region that connected West Africa to the rest of the continent.

Limitations of the analysis

The tools used in this study effectively model the technical and economic aspects of energy systems, with a particular focus on hydropower generation. While the results demonstrate the potential for within and between the power pool trading for economically efficient energy security, it did not model or address many of the political, institutional, and national security issues related to energy self-sufficiency. A few of the key issues that would inhibit implementation of the electricity trading options presented above are:

(1) The investment and maintenance of power transmission lines through nations that are not suppliers or demanders of the traded electricity
(2) The potential local pollution externalities in the electricity generating countries
(3) For hydropower, the allocation of water for high valued electricity to be consumed in foreign locations displacing local water use for household, irrigation, and environmental protection
(4) The vulnerability of national security due to limited energy self-sufficiency. With increasing electrification of African economies and the digitalization of many aspects of infrastructure, communications and daily life, depending on a significant portion of electricity supply come from a foreign source is just too great a national security and economic threat overshadowing the benefits of low cost and low carbon electricity supplies.

Conclusions and further research

In this study, we evaluated the impacts of 3 different electricity trade scenarios among four African power pools given a time series of available hydropower production. The available hydropower production was determined after all other higher priority water demands were met using a river basin model (WEAP). The objective of the power pool model (RTEP) was to maximize global welfare (i.e., meet total electricity demand at lowest cost including the cost of non-served energy) assuming perfect competition.

From the WEAP analysis, the CAPP (the Democratic Republic of Congo and more specifically the Inga hydropower projects) produced the most hydropower followed by the EAPP and SAPP and lastly the WAPP. This was similar in the RTEP analysis, which showed the connection of the two models. In the No Trade and Min Trade scenarios, there was a significant amount of hydropower production that went unused in all regions, especially in the CAPP and SAPP. A positive outcome of the Full Trade scenario was all hydropower was used; none was wasted.

By allowing trade among the power pools, the need to build new electricity production facilities was decreased by about 19 GW in the Min Trade scenario and an additional 10 GW in the Full Trade scenario. By reducing the need to expand production capacity, these regions saved money by not needing to build new facilities but instead were able to use electricity from other power pools.

There are three directions for this research to take to extend these findings

(1) Address the political economy issues of energy self-sufficiency in the modeling framework.
(2) Address Climate change impacts on the findings. Climate change will impact available hydropower production and will further impact the ability of regions to produce electricity and also impact the ability to trade electricity.
(3) Look at potential benefits of electricity trading on achieving regional SDG goals across a range of environmental sectors (e.g., GHG emission, air pollution, wetlands protection, instreamflow, land use, food production).

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix 1:. Base Electricity Demand

See Table A1.1, Table A1.2, Table A1.3, Table A1.4.

Appendix 2:. Electricity demand growth assumptions

See Table A2.1, Table A2.2, Table A2.3, Table A2.4.

Appendix 3:. Hydropower facilities

See Table A3.1.

References

[1] B. Bowen, F.T. Sparrow, Z. Yu, Modeling electricity trade policy for the twelve nations of the Southern African Power Pool (SAPP), Utilities Policy. 8 (1999) (1999) 183–197.
[2] Raff aello Cervigni, Rikard Liden, James E. Neumann, Kenneth M. Strzepek, Enhancing the Climate Resilience of Africa's Infrastructure: The Power and Water Sectors. Africa. Development Forum series. Washington, DC: World Bank. doi: 10.1596/978-1-4648-0466-3. License: Creative Commons Attribution CC BY 3.0 IGO, 2015.
[3] K. Lee, E. Brewer, C. Christiano, F. Meyo, E. Miguel, M. Podolsky, J. Rosa, C. Wolfram, Electrification for "Under Grid" households in Rural Kenya. Dev. Eng. 1 (2016) 26–35.
[4] Frederic Musisi, Connecting power across borders. The Niles. October 8, 2018.
[5] A. Rose, I. Perez-Arriaga, Technical Note: Proposed Model for Regional Power Sector Integration in Africa, 2016.
[6] Grania Rumbomboras, Africa's power pools can capture energy trade, ESI Africa. June 10, 2018.
[7] A.B. Sebitosi, R. Okou, Re-thinking the power transmission model for sub-Saharan Africa, Energy Pol. 38 (2010) (2010) 1448–1454.

[8] J. Sieber, D. Purkey, User Guide for WEAP21, Stockholm Environment Institute, 2007.
[9] S. Szabo, K. Bodis, T. Huld, M. Moner-Girona, Sustainable energy planning: Leapfrogging the energy poverty gap in Africa, Renew. Sustain. Energy Rev. 28 (2013) (2013) 500–509.
[10] P. Trotter, M. McManus, R. Maconachie, Electricity planning and implementation in sub-Saharan Africa: a systematic review, Renew. Sustain. Energy Rev. 74 (2017) (2017) 1189–1209.